

Digital Electronics

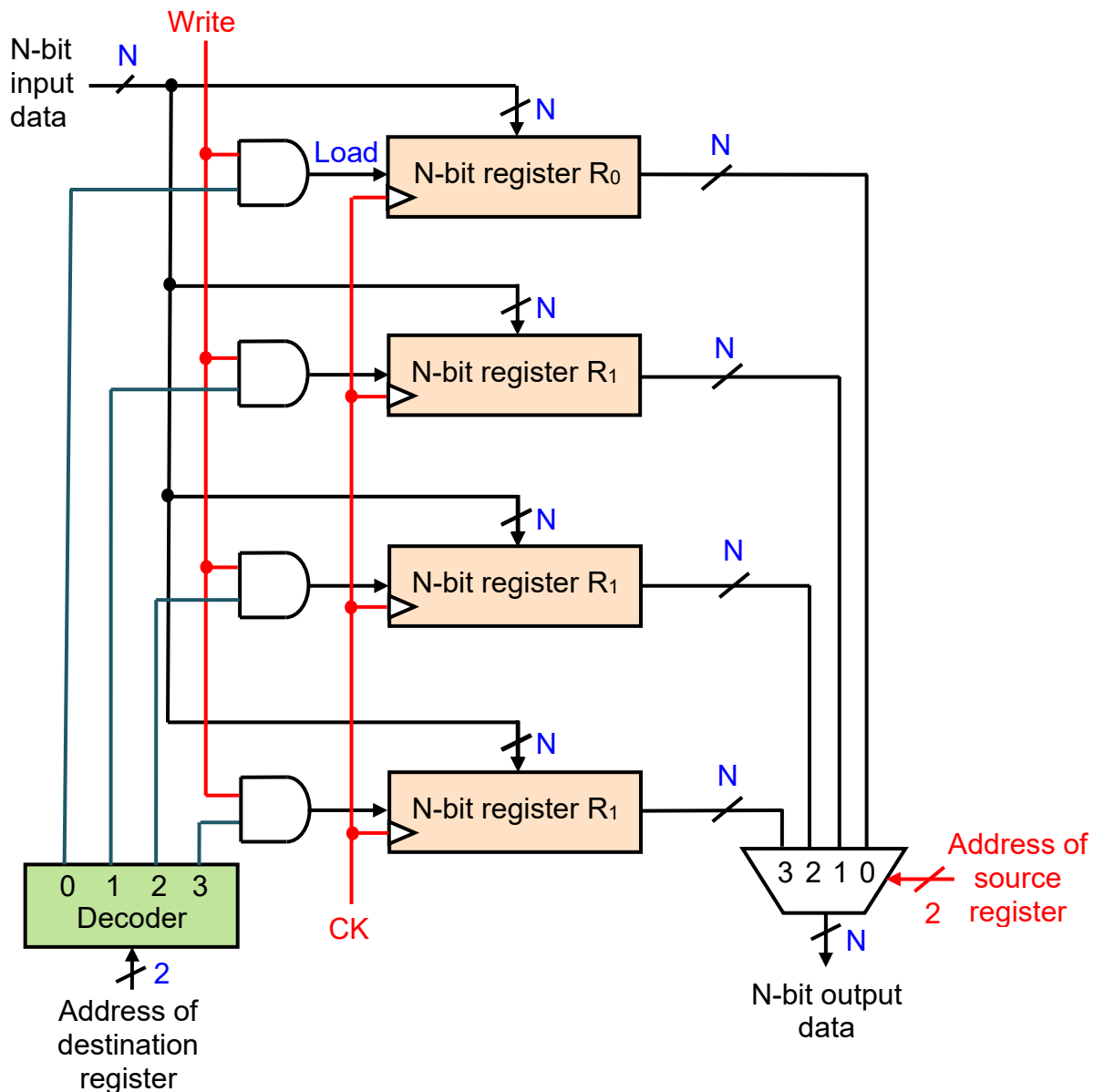
Circuit Example: Register File

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7. Register file with four registers

This is a memory circuit that can store four N-bit words. It is designed using four N-bit registers. This type of memory architecture is often referred to as a register file.



We can store (write) a binary word in this memory as follows: specify the address of the destination register using two bits (where do we want to store the word?) and set the Write input to 1 (to indicate that we want to store a word).

This will activate the Load input (Load = 1) for the specified register only. The binary word present at the N-bit input data will then be stored at the specified address on the next clock edge.

We can also read a binary word from this memory as follows: set the Write input to 0 (to indicate that we do not want to store a word) in order to deactivate the Load input (Load = 0) for all registers.

We also need to specify the address of the source register using two bits (at what address is located the word we want to access?). The word you want to access will then be available at the N-bit output data.